

Smart plant monitor

1. Introduction

I'm terrible at letting my plants die, so this project proposes an IoT based Smart Plant Monitor using Cisco Packet Tracer and Python. The system will simulate a soil moisture sensor that monitors the moisture level of a plant. When the moisture level falls below a predefined threshold, the system will trigger an alert message such as: "Water your plant!"

The project demonstrates how simulated IoT monitors can communicate with Python using network protocols. The aim is to create an example of environmental monitoring.

2. Proposed Technologies

- Humidity monitor in IoT, home devices, Cisco Packet Tracer.
- Raspberry Pi acting as an always-on listener/server
- Laptop running Cisco packet Tracer

3. Protocols

UDP for transmitting data

4. Programming Languages

Python for sensor listening and alert processing

5. System Design

The monitor in Packet Tracer will send simulated moisture readings over UDP to a Python listener application. The listener will process incoming data and trigger an alert if moisture levels fall below a threshold. In the future I would like an app to give me water my plant reminder.

6. Development Tools

- Visual Studio Code. This could be used because it includes built-in support for running python scripts and it helps manage multiple files such as sensor_listener.py and main.py
- Python
- Terminal / command line

7. Simulation and Platforms

- Cisco Packet Tracer
- Raspberry Pi

Deliverable

The final deliverable will be a working prototype demonstrating:

- Simulated soil moisture readings
- UDP communication between packet Tracer and Python
- Automated alert generation when moisture levels are too low.

FEEDBACK: Hi Edel. Thanks for that. If you have time, it might be worth using one of the IoT platforms we covered for monitoring/Dashboard, Good luck with it!